Data Mining

Chapter -2

Data mining

Data mining is the process of discovering interesting patterns and knowledge from large amounts of data.

The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.

Data Mining Applications

- **Insurance:** Data mining helps insurance companies to price their products profitable and deciding whether to approve policy applications, including risk modelling and management for prospective customers.
- **Education:** Data mining benefits educators to access student data, predict achievement levels and find students or groups of students which need extra attention. For example, students who are weak in maths subject.
- **Banking:** Data mining helps finance sector to get a view of market risks and manage regulatory compliance. It helps banks to identify probable defaulters to decide whether to issue credit cards, loans, etc. Bank and credit card companies use data mining tools to build financial risk models, detect fraudulent transactions and examine loan and credit applications.

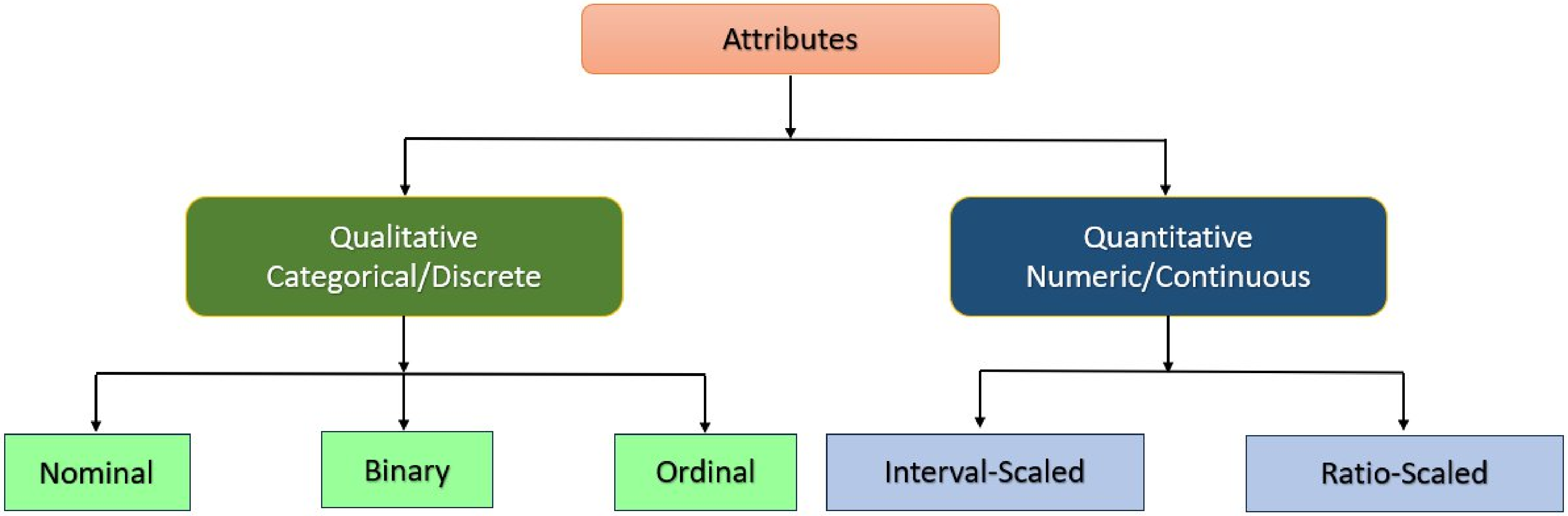
- **Retail:** Data Mining techniques help retail malls and grocery stores identify and arrange most sellable items in the most attentive positions. It helps store owners to comes up with the offer which encourages customers to increase their spending. Online retailers mine customer data and internet clickstream records to help them target marketing campaigns, ads and promotional offers to individual shoppers.

- **Service Providers:** Service providers like mobile phone and utility industries use Data Mining to predict the reasons when a customer leaves their company. They analyse billing details, customer service interactions, complaints made to the company to assign each customer a probability score and offers incentives.
- **E-Commerce:** E-commerce websites use Data Mining to offer cross-sells and up-sells through their websites. One of the most famous names is Amazon, who use Data mining techniques to get more customers into their eCommerce store.
- **Super Markets:** Data Mining allows supermarket's develop rules to predict if their shoppers were likely to be expecting. By evaluating their buying pattern, they could find woman customers who are most likely pregnant. They can start targeting products like baby powder, baby shop, diapers and so on.

- **Entertainment:** Streaming services do data mining to analyse what users are watching or listening to and to make personalized recommendations based on people's viewing and listening habits.
- **Healthcare:** Data mining helps doctors diagnose medical conditions, treat patients and analyse X-rays and other medical imaging results. Medical research also depends heavily on data mining, machine learning and other forms of analytics.

Attribute Types

- Attribute values are numbers or symbols assigned to an attribute.
- The type of the attribute can be determined based on the assigned value.
- The set of possible values - nominal, binary, ordinal, or numeric - the attribute can have.


Attributes
Qualitative
Categorical/Discrete
Quantitative
Numeric/Continuous
Nominal
Binary
Ordinal
Interval-Scaled
Ratio-Scaled

- **Nominal Attributes**
- The values of a nominal attribute are symbols or names of things. Each value represents some kind of category, code, or state.
- Nominal attributes are also referred to as Qualitative and Categorical attributes.

Attributes	Possible Values
hair_color	black, brown, red, green, and so on.
marital_status	single, married, divorced, and widowed.
occupation	teacher, doctor, farmer, student and so on.

Binary Attributes

- A binary attribute is a special nominal attribute with only two states: 0 or 1. Where 0 typically means that the attribute is absent, and 1 means that it is present.
- **Symmetric Binary Attribute**
- A binary attribute is symmetric if both of its states are equally valuable and carry the same weight.
- **Example:** the attribute gender having the states male and female.
- **Asymmetric Binary Attribute**
- A binary attribute is asymmetric if the outcomes of the states are not equally important.
- **Example:** Test results for COVID patient: Positive (1) and Negative (0).
- By convention, we code the most important outcome, which is usually the rarest one, by 1 (e.g., COVID positive) and the other by 0 (e.g., COVID negative).

Ordinal Attributes:

- An ordinal attribute is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.
- Ordinal attributes are also referred to as Qualitative and Categorical attributes.
- **Example:** An ordinal attribute drink_size corresponds to the size of drinks available at a fast-food restaurant.
- This attribute has three possible values: small, medium, and large.
- The values have a meaningful sequence (which corresponds to increasing drink size);
- However, we cannot tell from the values how much bigger, say, a medium is than a large.

- Ordinal attributes are useful in surveys, In one survey, participants were asked to rate how satisfied they were as customers.
- Customer satisfaction had the following ordinal categories:

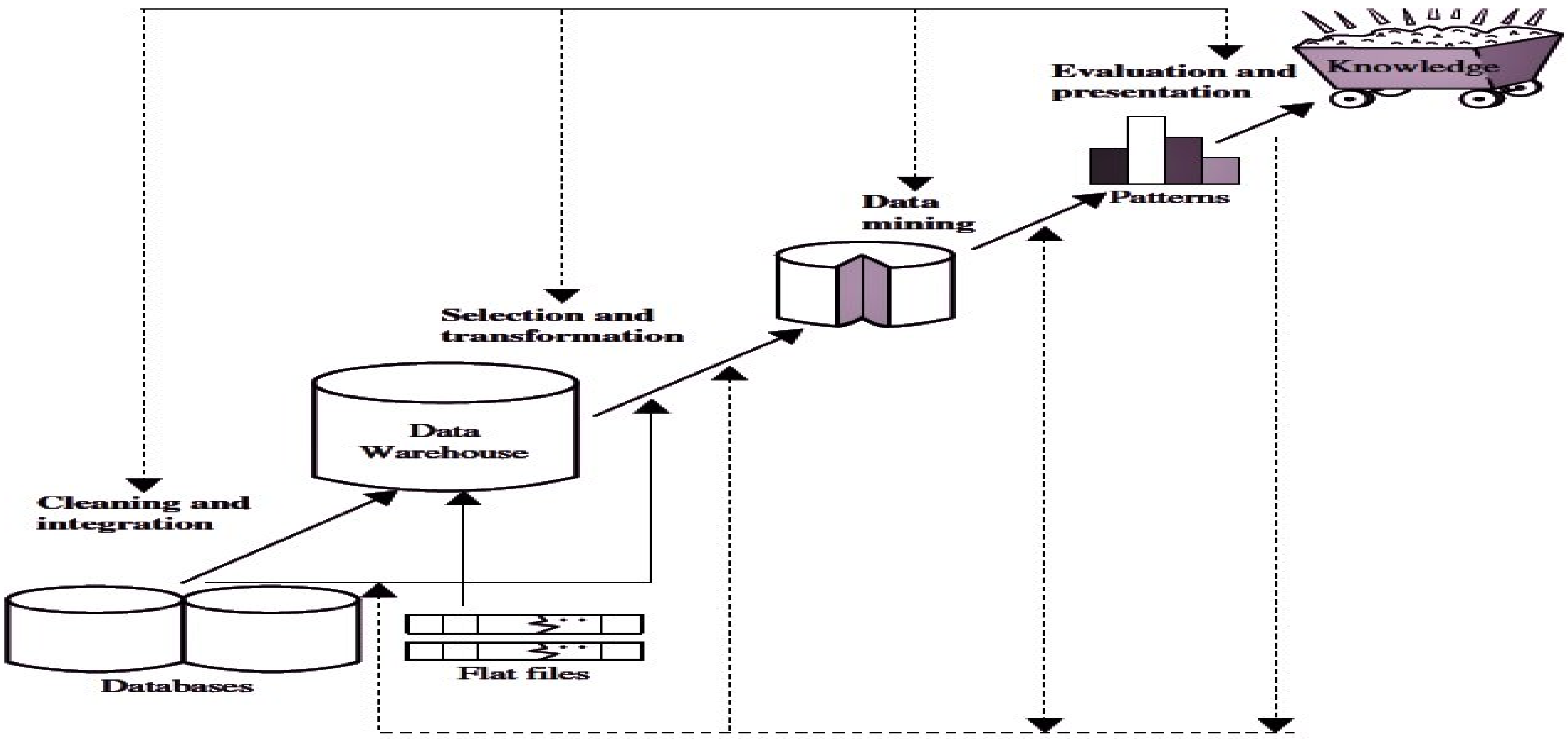
0:very dissatisfied
1: somewhat dissatisfied
2: neutral
3: satisfied
4: very satisfied.

Knowledge Discovery from Data (KDD)

- The need of data mining is to extract useful information from large datasets and use it to make predictions or better decision-making.
- Nowadays, data mining is used in almost all places where a large amount of data is stored and processed.
- **For examples:** Banking sector, Market Basket Analysis, Network Intrusion Detection.
- Data Mining also known as **K**nowledge **D**iscovery from **D**ata or **KDD**.

- **Knowledge Discovery from Data (KDD) Process**
- KDD is a process that involves the extraction of useful, previously unknown, and potentially valuable information from large datasets.
- The KDD process is an iterative process and it requires multiple iterations of the above steps to extract accurate knowledge from the data.

KDD


Knowledge
Evaluation and presentation
Patterns
Data mining
Selection and transformation
Data Warehouse
Cleaning and integration
Databases
Flat files

- The following steps are included in KDD process:

1. **Data Cleaning**
2. **Data Integration**
3. **Data Selection**
4. **Data Transformation**
5. **Data Mining**
6. **Pattern Evaluation**
7. **Knowledge Representation**

- **Data Cleaning**
- Data cleaning is defined as removal of **noisy** and **irrelevant/ inconsistent** data from data collection.
- Cleaning in case of **Missing values**.
- Cleaning **noisy data**, where noise is a **random** or **variance error**.
- In this step, the **noise** and **inconsistent** data is removed.

Data Integration

- Data integration is defined as heterogeneous **data from multiple data sources** combined in a common source (Data Warehouse).
- i.e., In this step, multiple data sources may be combined as single data source.
- A popular trend in the information industry is to perform **data cleaning** and **data integration** as a **data preprocessing** step, where the resulting data are stored in a **data warehouse**.

Data Selection

- Data selection is defined as the process where **data relevant to the analysis** is decided and retrieved from the data collection.
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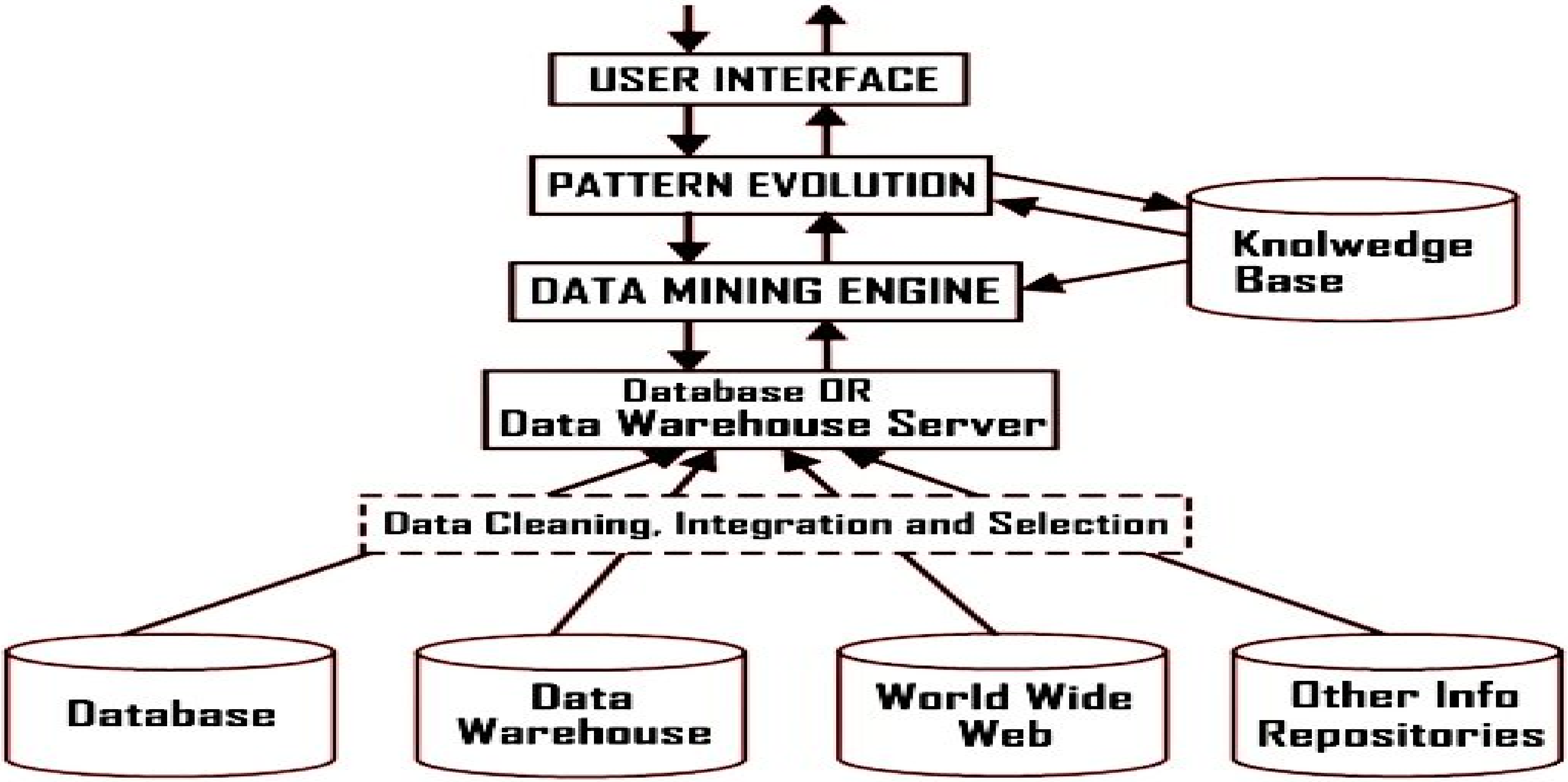
- This step in the KDD process is **identifying** and **selecting** the relevant data for analysis.

- **Data Transformation**
- Data Transformation is defined as the process of **transforming data into appropriate form** required by mining procedure.
- This step involves reducing the data dimensionality, aggregating the data, normalizing it, and discretizing it to prepare it for further analysis.
- **Data Mining**
- This is the heart of the KDD process and involves applying **various data mining techniques** to the transformed data to discover **hidden patterns, trends, relationships**, and **insights**. A few of the most common data mining techniques include clustering, classification, association rule mining, and anomaly detection.

- **Pattern Evaluation**
- After the data mining, the next step is to evaluate the **discovered patterns** to determine their usefulness and relevance.
- This involves assessing the quality of the patterns, evaluating their significance, and selecting the most promising patterns for further analysis

- **Knowledge Representation**
- This step involves **representing the knowledge** extracted from the data in a way humans can easily understand and use.
- This can be done through visualizations, reports, or other forms of communication that provide meaningful insights into the data

- Data mining is a very important process where potentially useful and previously unknown information is extracted from large volumes of data. There are several components involved in the data mining process.
- The major components of any data mining system are data source, data warehouse server, data mining engine, pattern evaluation module, graphical user interface and knowledge base.


USER INTERFACE
PATTERN EVOLUTION
Knolwedge Base
DATA MINING ENGINE
Database OR
Data Warehouse Server
Data Cleaning, Integration and Selection
Database
Data Warehouse
World Wide Web
Other Info Repositories

Data Sources

- **Database, data warehouse, World Wide Web (WWW), text files** and **other documents** are the actual sources of data. You need large volumes of historical data for data mining to be successful.
- Organizations usually store data in databases or data warehouses. Data warehouses may contain one or more databases, text files, spreadsheets, or other kinds of information repositories. Sometimes, data may reside even in plain text files or spreadsheets. World Wide Web or the Internet is another big source of data.
- The data needs to be **cleaned, integrated,** and **selected** before passing it to the database or data warehouse server. As the data is from different sources and in different formats, it cannot be used directly for the data mining process because the data might not be complete and reliable. So, first data needs to be cleaned and integrated

- **Database or Data Warehouse Server**
- The database or data warehouse server contains the **actual data** that is **ready to be processed**. Hence, the server is responsible for retrieving the relevant data based on the data mining request of the user.

- **Data Mining Engine**
- The data mining engine is the **core component** of any data mining system. It consists of **several modules** for performing data mining tasks including association, classification, characterization, clustering, prediction, time-series analysis etc.
- **Pattern Evaluation Modules**
- The pattern evaluation module is mainly responsible for the **measure of interestingness of the pattern** by using a **threshold value**. It interacts with the data mining engine to focus the search towards interesting patterns.
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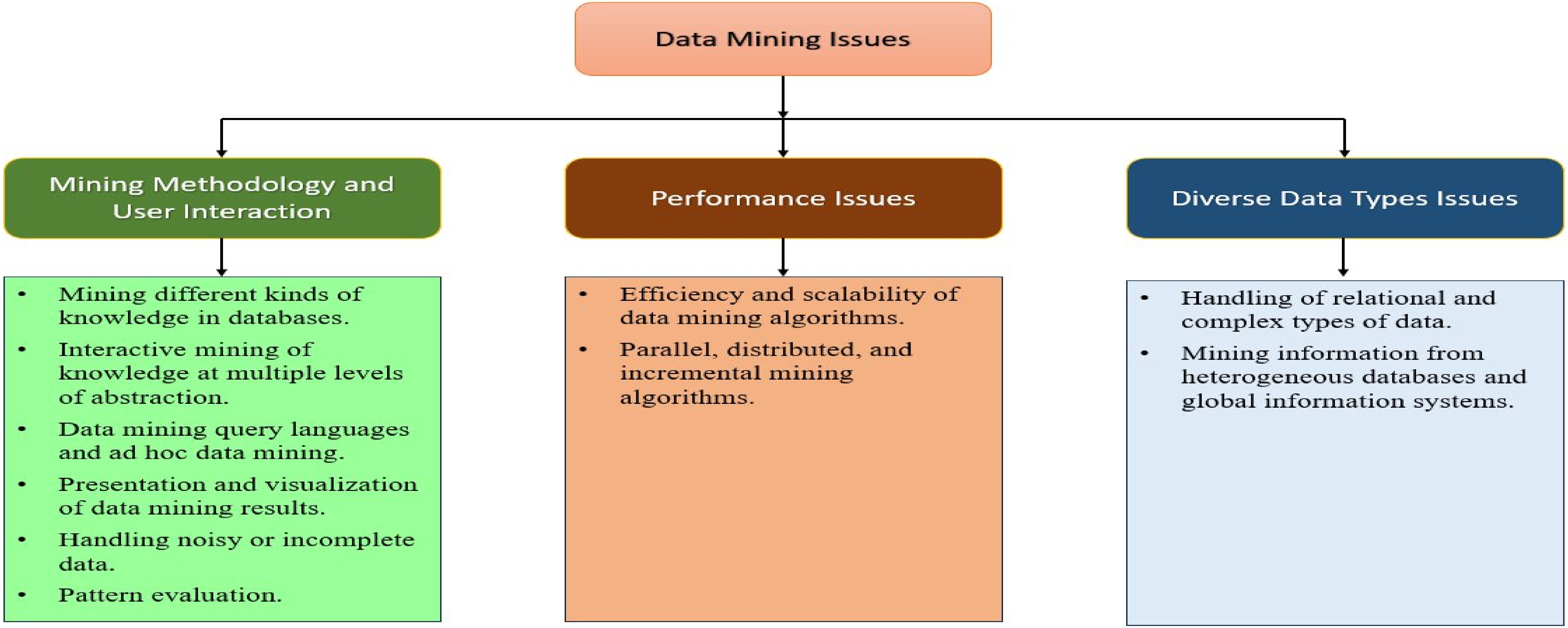
- **Graphical User Interface**
- The graphical user interface module provides the **communication** between the **user** and the **data mining system**. This module helps the user use the system easily
- and efficiently without knowing the real complexity behind the process.
- When the user specifies a query or a task, this module interacts with the data mining system and displays the result in an easily understandable manner.
- **Knowledge Base**
- The knowledge base is **helpful** in the **whole data mining process**. It might be useful for guiding the search or evaluating the interestingness of the result patterns. The knowledge base might even contain user beliefs and data from user experiences that can be useful in the process of data mining.
- The data mining engine might get inputs from the knowledge base to make the **result more accurate** and **reliable**. The pattern evaluation module interacts with the knowledge base on a regular basis to get inputs and also to update it.

Issues in Data Mining

- **Major issues in Data Mining**
- Data mining, the process of extracting knowledge from data, has become increasingly important as the amount of data generated by individuals, organizations, and machines has grown exponentially.Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources.

- The above factors may lead to some issues in data mining. These issues are mainly divided into three categories, which are given below:

1. **Mining Methodology and User Interaction**
2. **Performance Issues**
3. **Diverse Data Types Issues**


Data Mining Issues
Mining Methodology and User Interaction
Performance Issues
Diverse Data Types Issues
Mining different kinds of knowledge in databases.
Interactive mining of knowledge at multiple levels of abstraction.
Data mining query languages and ad hoc data mining.
Presentation and visualization of data mining results.
Handling noisy or incomplete data.
Pattern evaluation.
Efficiency and scalability of data mining algorithms.
Parallel, distributed, and incremental mining algorithms.
Handling of relational and complex types of data.
Mining information from heterogeneous databases and global information systems.

- **Mining Methodology and User Interaction**
- It refers to the following kinds of issues
- **Mining different kinds of knowledge in databases** − Different users may be interested in different kinds of knowledge. Therefore, it is necessary for data mining to cover a broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction** − The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- **Data mining query languages and ad hoc data mining** − Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.

- **Presentation and visualization of data mining results** − Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- **Handling noisy or incomplete data** − The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation** − The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

Performance Issues

- There can be performance-related issues such as follows
- **Efficiency and scalability of data mining algorithms** − In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms** − The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion. Theincremental algorithms, update databases without mining the data again from scratch.

- **Diverse Data Types Issues**
- **Handling of relational and complex types of data** – The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kinds of data.
- **Mining information from heterogeneous databases and global information systems** – The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore, mining the knowledge from them adds challenges to data mining.

1. **Poor Data Quality** – Data may have mistakes, missing values, or duplicates.
2. **Data Integration** – It is difficult to combine data from many sources.
3. **Scalability** – Very large data sets need powerful tools and storage.
4. **Privacy and Security** – Personal or sensitive data must be protected.
5. **Choosing the Right Method** – Picking the best mining technique is not always easy.
6. **Understanding Results** – Patterns and outcomes may be hard to explain.
7. **Changing Data** – Data is not always stable; it changes with time.

- **Data Preprocessing**

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What is Data Preprocessing?

- Data preprocessing is a crucial step in data mining. It involves transforming raw data into a clean, structured, and suitable format for mining. Proper data preprocessing helps improve the quality of the data, enhances the performance of algorithms, and ensures more accurate and reliable results.
- **Why Preprocess the Data?**
- In the real world, many databases and data warehouses have **noisy**, **missing**, and **inconsistent** data due to their huge size. Low quality data leads to low quality data mining.

- **Noisy:** Containing errors or outliers. **E.g., Salary = “-10”**
- Noisy data may come from
- Human or computer error at data entry.
- Errors in data transmission.
- **Missing:** lacking certain attribute values or containing only aggregate data. **E.g., Occupation = “”**
- Missing (Incomplete) may data come from
- “Not applicable” data value when collected.
- Human/hardware/software problems.

- **Inconsistent:** Data inconsistency meaning is that different versions of the same data appear in different places.For **example**, the ZIP code is saved in one table as **1234-567** numeric data format; while in another table it may be represented in **1234567**.
- Inconsistent data may come from
- Errors in data entry.
- Merging data from different sources with varying formats.
- Differences in the data collection process.
- Data preprocessing is used to improve the quality of data and mining results. And The goal of data preprocessing is to enhance the accuracy, efficiency, and reliability of data mining algorithms

- Data preprocessing is an essentialstepin the knowledge discovery process, because quality decisions must be based on qualitydata.And Data Preprocessing involvesData Cleaning, Data Integration, Data Reduction and Data Transformation.
- **Steps in Data Preprocessing**
- **1. Data Cleaning**
- Data cleaning is a process that "cleans" the data by filling in the missing values, smoothing noisy data, analyzing, and removing outliers, and removing inconsistencies in the data.

- If usersbelieve the data are dirty, they are unlikely to trust the results of any data mining that hasbeen applied.
- Real-world data tend to be incomplete, noisy, and inconsistent. Data cleaning (or datacleansing) routines attempt to fill in missing values, smooth out noise while identifyingoutliers, and correct inconsistencies in the data.

- Imagine that you need to analyze All Electronics sales and customer data. You note thatmany tuples have no recorded value for several attributes such as customer income. Howcan you go about filling in the missing values for this attribute? There are several methods to fill the missing values.
- Those are,

a. **Ignore the tuple:** This is usually done when the class label is missing(classification). This method is not very effective, unless the tuple contains several attributes with missing values.

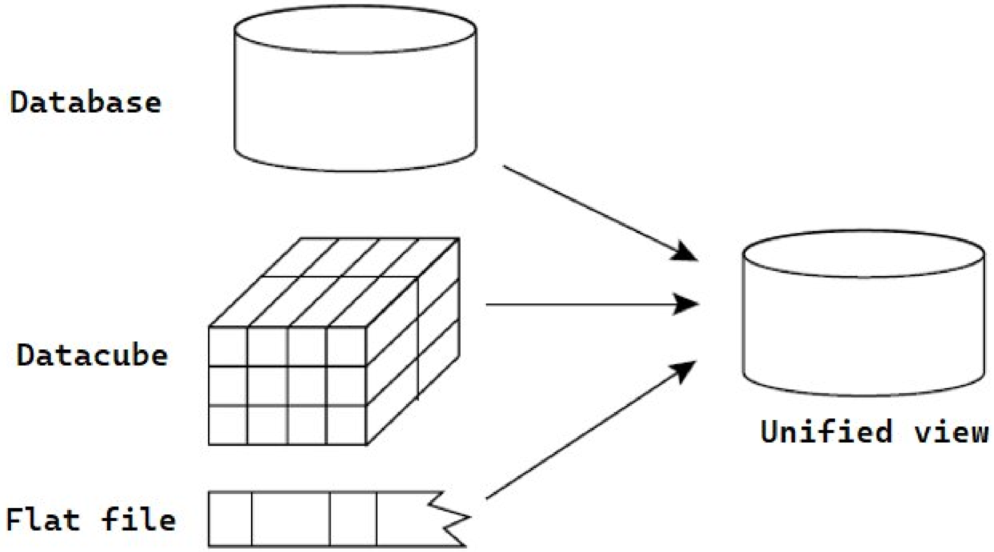
b. **Fill in the missing value manually:** In general, this approach is time consuming andmay not be feasible given a large data set with many missing values.

c. **Use a global constant to fill in the missing value:** Replace all missing attribute valuesby the same constant such as a label like "Unknown" or "- ∞ ".

d. **Use the attribute mean or median to fill in the missing value:** Replace all missing values in the attribute by the mean or median of that attribute values.

. Data Integration

- Data integration is the process of combining data from multiple sources into a single, unified view. This process involves identifying and accessing the different data sources, mapping the data to a common format. Different data sources may include multiple data cubes, databases, or flat files.
- The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems or platforms, in order to gain a more complete and accurate understanding of the data.
- Data integration strategy is typically described using a triple (G, S, M) approach, where G denotes the global schema, S denotes the schema of the heterogeneous data sources, and M represents the mapping between the queries of the source and global schema


Database
Datacube
Flat file
Unified view

- **Example:** To understand the (G, S, M) approach, let us consider a data integration scenario that aims to combine employee data from two different HR databases, database A and database B. The global schema (G) would define the unified view of employee data, including attributes like EmployeeID, Name, Department, and Salary.
- In the schema of heterogeneous sources, database A (S1) might have attributes like EmpID, FullName, Dept, and Pay, while database B's schema (S2) might have attributes like ID, EmployeeName, DepartmentName, and Wage. The mappings (M) would then define how the attributes in S1 and S2 map to the attributes in G, allowing for the integration of employee data from both systems into the global schema.

- **Issues in Data Integration**
- There are several issues that can arise when integrating data from multiple sources, including:

a. Data Quality: Data from different sources may have varying levels of accuracy, completeness, and consistency, which can lead to data quality issues in the integrated data.

b. Data Semantics: Integrating data from different sources can be challenging because the same data element may have different meanings across sources.

c. Data Heterogeneity: Different sources may use different data formats, structures, or schemas, making it difficult to combine and analyze the data.

- **Data Reduction**
- Imagine that you have selected data from the AllElectronics data warehouse for analysis.The data set will likely be huge! Complex data analysis and mining on huge amounts ofdata can take a long time, making such analysis impractical or infeasible.
- Data reduction techniques can be applied to obtain a reduced representation of thedata set that ismuch smaller in volume, yet closely maintains the integrity of the originaldata. That is, mining on the reduced data set should be more efficient yet produce thesame (or almost the same) analytical results.
- In simple words,Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information.
- This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.

- There are several different data reduction techniques that can be used in data mining, including:

a. Data Sampling: This technique involves selecting a subset of the data to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.

b. Dimensionality Reduction: This technique involves reducing the number of features in the dataset, either by removing features that are not relevant or by combining multiple features into a single feature.

c. Data compression:This is the process of altering, encoding, or transforming the structure of data in order to save space. By reducing duplication and encoding data in binary form, data compression creates a compact representation of information. And it involves the techniques such as lossy or lossless compression to reduce the size of a dataset.

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Data Transformation

- Data transformation in data mining refers to the process of converting raw data into a format that is suitable for analysis and modelling.
- The goal of data transformation is to prepare the data for data mining so that it can be used to extract useful insights and knowledge.

Aggregation

Year 2008

Quarter	Sales
Q1	$224,000
Q2	$408,000
Q3	$350,000
Q4	$586,000

Year	Sales
2008	$1,568,000
2009	$2,356,000
2010	$3,594,000

Normalization

$-2, 32, 100, 59, 48 \longrightarrow -0.02, 0.32, 1.00, 0.59, 0.48$

- Data transformation typically involves several steps, including:

1. **Smoothing:** It is a process that is used to remove noise from the dataset using techniques include binning,regression, and clustering.
2. **Attribute construction (or feature construction):** In this, new attributes are constructed and added from the given set of attributes to help the mining process.
3. **Aggregation:** In this, summary or aggregation operations are applied to the data. Forexample, the daily sales data may be aggregated to compute monthly and annual total amounts.
4. **Data normalization:** This process involves converting all data variables into a small range.such as -1.0 to 1.0, or 0.0 to 1.0.
5. **Generalization:** It converts low-level data attributes to high-level data attributes using concept hierarchy. For Example, Age initially in Numerical form (22,) is converted into categorical value (young, old).

Integration and Transformation in Pre processing

1.Data Integration

1. Data often comes from different sources like databases, files, or websites.
2. Integration means **combining this data into one single view**.
3. It helps remove duplicates and inconsistencies so that the data is clean and uniform.
4. Example: Merging customer records from online and offline stores.

2.Data Transformation

1. After integration, the data may not be in the right format.
2. Transformation means **converting data into a proper format or structure** for mining.
3. It includes normalization (scaling numbers), aggregation (summarizing data), and generalization (replacing details with higher-level concepts).
4. Example: Converting prices into the same currency or normalizing age values.

Method Name	Irregularity	Output
Data Cleaning	Missing, Nosie, and Inconsistent data	Quality Data before Integration
Data Integration	Different data sources (data cubes, databases, or flat files)	Unified view
Data Reduction	Huge amounts of data can take a long time, making such analysis impractical or infeasible.	Reduce the size of a dataset and maintains the integrity.
Data Transformation	Raw data	Prepare the data for data mining